
Differential plasma-wave self-phase modulation from local ZZ interactions in a spin chain

Computational research report

Abstract

A longitudinal spin interaction does not by itself establish a correspondence with nonlinear plasma dynamics: it changes the vacuum dispersion as well as the interaction between excitations. We isolate these contributions for a staggered spin chain with attractive ZZ couplings between sites two lattice spacings apart. The coupling is local on each staggered field component and generates the cubic term of the weakly relativistic electromagnetic plasma-envelope equation after band projection and a mean-field closure. Full 12-qubit statevector evolution reproduces the differential nonlinear carrier-frequency downshift of an independently integrated nonlinear Schrödinger equation with errors of 0.42–1.47% across an eightfold coupling range, below a pre-specified 15% tolerance. A weak spatial modulation gives 3.42% error at a fixed interior readout. The comparison requires both a dilute calibration and subtraction of the unextended spin chain’s finite-density phase. Stronger coupling and longer evolution fail, with errors of 22.37% and 71.74%, respectively. An explicit finite-spin coherence formula explains the onset of failure. These results establish a calibrated, short-time nonlinear spectroscopy correspondence, while exposing why phase agreement on a small spin lattice does not establish nonlinear transport or quantum advantage.

1 Nonlinear plasma spectroscopy with a local spin model

The relativistic increase in electron inertia lowers the frequency of an electromagnetic plasma wave as its intensity increases. In a slowly varying envelope this effect produces self-phase modulation, the local nonlinear response that also participates in focusing and modulational dynamics. It offers a sharper first test of a quantum plasma simulator than the presence of a formally interacting Hamiltonian: the interaction strength must predict a measurable, intensity-dependent frequency shift.

A staggered spin chain can reproduce the linear dispersion and scattering of plasma waves using shallow local circuits [Sundar et al., 2026]. Extending that construction beyond a single excitation raises two problems. A ZZ bond contains an effective one-body field as well as a density interaction. Moreover, spin one-half excitations are hard-core, so even the unextended chain need not have the coherent-field dynamics of unrestricted free bosons. Agreement of a nonlinear observable must distinguish these effects.

We study a fixed, component-local extension and a differential Ramsey measurement that performs this separation. The plasma reference is the focusing envelope equation associated with the weakly relativistic response of a cold, circularly polarized wave [Akhiezer and Polovin, 1956, Siminos et al., 2014]. Our contribution is the explicit connection between the staggered coupling graph, its calibrated nonlinear phase coefficient, and the finite-time range over which full spin dynamics agrees with that coefficient. We also test a spatially modulated input and identify failures caused by correlations and boundaries. The field-theory closure is approximate; no nonlinear evolution law is imposed on the quantum statevector.

2 From ZZ bonds to the plasma envelope

2.1 Hamiltonian and component locality

For an open chain, keep the original chiral exchange and staggered field,

$$H_0 = -\frac{J}{4} \sum_{j=1}^{N-1} (X_j Y_{j+1} - Y_j X_{j+1}) + \frac{1}{2} \sum_{j=1}^N \Delta_j (-1)^j Z_j, \quad (1)$$

$$H(\epsilon) = H_0 - \frac{\epsilon}{4} \sum_{j=1}^{N-2} Z_j Z_{j+2}, \quad K_{ij} = -\frac{\epsilon}{4} \mathbf{1}_{j-i=2} \quad (i < j). \quad (2)$$

Here $H_{ZZ} = \sum_{i < j} K_{ij} Z_i Z_j$; all other matrix entries vanish. No exchange term, staggered field, or boundary bond of H_0 is changed. Throughout, $J = \hbar = a = 1$ unless units are restored. We use standard Pauli matrices, $Z|0\rangle = |0\rangle$, and one-based physical site indices. Thus $n_j = (1 - Z_j)/2$ and $a_j = |0\rangle_j \langle 1|$; the one-particle mass is $m_j = -\Delta_j (-1)^j$. A scalar vacuum energy is removed in both numerical implementations.

The staggered representation places each component of the underlying two-component field on one parity sublattice. Coupling j to $j + 2$ therefore links adjacent samples of the *same* component, separated by $2a$. It is a finite-range discretization of a local quartic energy. A bond joining opposite parity components would instead primarily couple the positive- and negative-frequency components; their overlap is small near the positive band edge used here. The chosen graph is the same at every coupling and at every system size.

The exact operator identity

$$-\frac{\epsilon}{4} Z_i Z_j = -\epsilon n_i n_j + \frac{\epsilon}{2} (n_i + n_j) - \frac{\epsilon}{4} \quad (3)$$

exhibits both contributions. In the interior the coordination number is $z_j = 2$, so the vacuum-referenced one-body shift is ϵ ; the two ends of each component have $z_j = 1$. The attractive density term is $-\epsilon n_j n_{j+2}$. For an approximately uniform component density ρ , its energy per component site is $-\epsilon \rho^2$, with chemical potential $-2\epsilon \rho$. Equation (2) also proves $H(0) = H_0$ exactly.

2.2 Classical plasma equation and coefficient identification

For circular polarization, immobile ions and negligible leading density perturbation, the normalized transverse vector potential $\mathcal{A}$ satisfies the cold relativistic wave model

$$\left[\partial_t^2 - c^2 \partial_x^2 + \frac{\omega_p^2}{\sqrt{1 + |\mathcal{A}|^2}} \right] \mathcal{A} = 0. \quad (4)$$

The nonlinear inertia in this model is the origin of the focusing cubic envelope term; higher-order relativistic and density-response corrections are analyzed in the plasma literature [Akhiezer and Polovin, 1956, Siminos et al., 2014]. To display our normalization, set $c = Ja$, $\omega_p = \Delta$ and $\mathcal{A} = \alpha \psi(x, t) e^{-i\Delta t}$. Expanding the inverse Lorentz factor and neglecting $\partial_t^2 \psi$ relative to $\Delta \partial_t \psi$ gives

$$i \partial_t \psi = -D \partial_x^2 \psi - g |\psi|^2 \psi, \quad D = \frac{J^2 a^2}{2\Delta}, \quad g = \frac{\Delta \alpha^2}{4} = 2\epsilon, \quad \alpha^2 = \frac{8\epsilon}{\Delta}. \quad (5)$$

This is the leading focusing limit of the perturbed plasma NLSE, with an explicitly stated field normalization. Varying ϵ at fixed spin density varies the physical quiver intensity in this encoding. It does not imply that a plasma's material coefficient can be changed without changing that encoding.

For a uniform input $\psi(x, 0) = \sqrt{\rho}$, the equation has

$$\psi(x, t) = \sqrt{\rho} e^{2i\epsilon \rho t}, \quad \delta\omega = -2\epsilon \rho, \quad R_{\text{PDE}} \equiv -\delta\omega = 2\epsilon \rho. \quad (6)$$

The reference fields are nevertheless obtained by numerical PDE evolution, with Eq. (6) serving as an independent solver check. The observable is a frequency downshift, not an instability growth rate.

2.3 What the spin-to-field reduction assumes

The exact Heisenberg equation is

$$i\partial_t a_j = m_j a_j + (1 - 2n_j) \sum_l h_{jl}^{\text{hop}} a_l + \left(\frac{\epsilon z_j}{2} - \epsilon \sum_{|l-j|=2} n_l \right) a_j, \quad (7)$$

where $h_{j,j+1}^{\text{hop}} = iJ/2$ and $h_{j,j-1}^{\text{hop}} = -iJ/2$. The last term is the exact source of the desired nonlinear phase. Factorizing $\langle n_l a_j \rangle \simeq \langle n_l \rangle \langle a_j \rangle$ and replacing the two neighboring densities by $\rho(x)$ yields $-2\epsilon\rho\langle a_j \rangle$. The interaction energy gives the same coefficient without any fit.

The linear bulk bands are $E_{\pm}(k) = \pm\sqrt{\Delta^2 + J^2 \sin^2(ka)}$. Near $k = 0$, the positive band resides predominantly on odd sites and $E_+(k) - \Delta = Dk^2 + \mathcal{O}(k^4)$. Removing the bulk vacuum shift and projecting to this component leads to Eq. (5). The classical amplitude is defined by population and phase, $\psi = \sqrt{\rho}e^{i\theta}$; a spin coherent state has $\langle a \rangle = \sqrt{\rho(1-\rho)}e^{i\theta}$, so its squared coherence must not be substituted for ρ .

These are the usual dilute spin-to-boson and classical closure ideas, related to the Holstein–Primakoff construction [Holstein and Primakoff, 1940], rather than a new exact bosonization. The reduction omits connected correlations, saturation in the exchange term, higher gradients, negative-band admixture and endpoint coordination effects. At finite spin one-half there is no large-occupation classical limit on a single site. We therefore test the *incremental* ZZ response experimentally in the numerical model, and retain all omitted effects in the full statevector. In particular, subtraction of the finite-density H_0 response is part of the correspondence we establish.

3 A fixed differential spectroscopy protocol

The main simulations use $N = 12$, $\Delta = 8$, and six populated odd sites. The initial state is

$$|\Psi_{\rho}\rangle = \bigotimes_{j \text{ odd}} \left(\sqrt{1-\rho}|0\rangle_j + \sqrt{\rho}|1\rangle_j \right) \bigotimes_{j \text{ even}} |0\rangle_j, \quad \rho = 0.15. \quad (8)$$

Its mean excitation number is 0.9 and its weight in sectors with at least two excitations is 22.35%. Every evolution uses all 4096 amplitudes. The primary couplings are $\epsilon/J = 0.05, 0.1, 0.2, 0.4$; the additional pre-declared stress points are 0.8, 1.6. The relative-error criterion is 15% at each primary point. The graph, couplings, readout, time interval and tolerance were fixed in a timestamped method record before any validation run.

Let $S_{\epsilon}(t)$ be the mean of $\langle a_5 \rangle$ and $\langle a_7 \rangle$ in Eq. (8), divided by its initial value. Define $L_{\epsilon}(t)$ analogously for the dilute Ramsey state

$$|\Psi_d\rangle = \frac{1}{\sqrt{2}} \left[|\text{vac}\rangle + \sum_{j \text{ odd}} f_j |j\rangle \right], \quad f_j = \frac{1}{\sqrt{6}}. \quad (9)$$

For modulated inputs, f_j is proportional to the specified initial $\sqrt{\rho_j}$, normalized in the one-excitation sector. This calibration is also evolved as a full statevector. The fixed measurement is

$$\Phi_{\epsilon}(t) = \text{unwrap arg} \left[\frac{S_{\epsilon}(t)/L_{\epsilon}(t)}{S_0(t)/L_0(t)} \right], \quad R_{\text{spin}} = \text{slope}_{[0,T]} \Phi_{\epsilon}(t), \quad T = 1/J. \quad (10)$$

All 101 samples, separated by $0.01/J$, enter an ordinary least-squares fit with an intercept. No early-time derivative is substituted for a measured trajectory, and the interval is not optimized after a run. L_{ϵ} removes the one-body ZZ shift and the linear lattice response; S_0/L_0 removes the hard-core phase of the unextended model. The zero-coupling controls are shared across a given sweep, so the rows do not represent statistically independent calibration experiments.

The spin solver uses a second-order diagonal/odd/even Suzuki splitting with $\delta = 0.005/J$. A smaller step than the linear demonstration is necessary because the nonlinear experiment has a much larger gap and resolves small differential phases. We compare it with $0.01/J$, $0.0025/J$ and an independent sparse matrix-exponential action on the full Hilbert space. The PDE solver uses 128 Fourier points

Table 1: Differential frequency downshift from full spin evolution and the independent plasma PDE. All declared couplings are shown. The linear statistic η_{lin} is the maximum nonzero-mode error against the exact nine-site spectrum at $\epsilon = 0$, using the disclosed refined Ramsey step $0.05/J$. It is a *shared zero-coupling control*, repeated for clarity, not a finite- ϵ spectrum. All nine modes in both mass cases pass; the zero-frequency mode is checked absolutely. $H(0)$ and H_0 give identical statevectors. The nonlinear tolerance is 15%.

ϵ/J	R_{spin}/J	R_{PDE}/J	Error (%)	η_{lin} (%)	< 5%	Regime
0	0.000000	0.000000	—	0.934	pass	$H(0) = H_0$
0.05	0.014937	0.015000	0.42	0.934	pass	primary
0.1	0.029869	0.030000	0.44	0.934	pass	primary
0.2	0.059639	0.060000	0.60	0.934	pass	primary
0.4	0.118240	0.120000	1.47	0.934	pass	primary
0.8	0.227138	0.240000	5.36	0.934	pass	stress, pass
1.6	0.372638	0.480000	22.37	0.934	pass	stress, fail

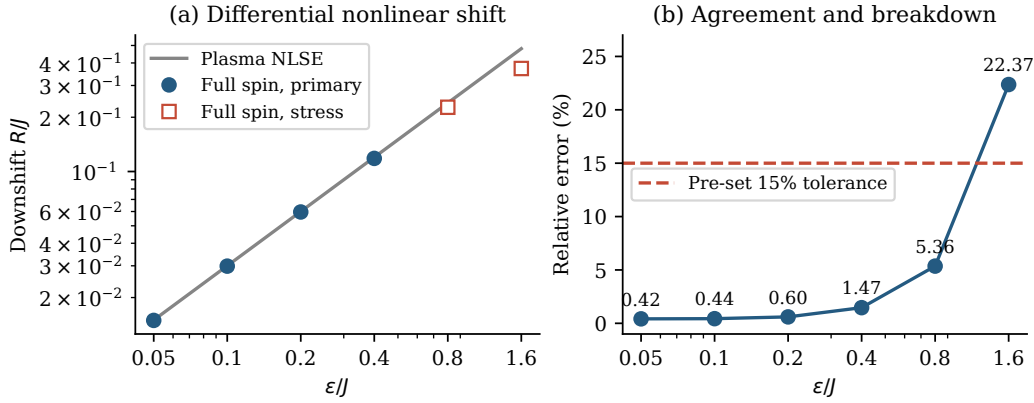


Figure 1: Measured differential downshift and the pre-specified error criterion. Filled circles mark the primary range; open squares mark stronger-coupling tests. The reference curve is computed by PDE evolution with the coefficient fixed by Eq. (3). No coefficient is fitted to the spin results.

on a periodic interval of length $12a$ and a split-step interval $0.005/J$. All PDE reference fields were saved before evaluating spin residuals. A 256-point, half-step solution checks the nonuniform case; the periodic PDE is an interior reference for the open spin chain, not a claim of matching boundary conditions.

4 Quantitative correspondence and its linear controls

4.1 The nonlinear shift across the coupling sweep

Table 1 and Fig. 1 report every declared coupling. The primary errors range from 0.42% to 1.47%, comfortably below the fixed tolerance. The response increases with the expected attractive-interaction sign. The first stress point still agrees within 5.36%; the strongest misses the tolerance with 22.37% error.

The distinction between absolute and differential phase is substantial. At $\epsilon = 0.4J$, the dilute-calibrated spin phase has slope $0.108880J$, while H_0 contributes $-0.009360J$; subtracting the latter gives the value in Table 1. Figure 2 shows both measured traces. At the weakest primary coupling, omitting this subtraction would produce 62.82% error relative to the NLSE shift. Consequently the data validate the incremental quartic response, not an uncalibrated equality of the spin and classical fields. The H_0 control lacks that additional attractive response, although it has its own hard-core contribution.

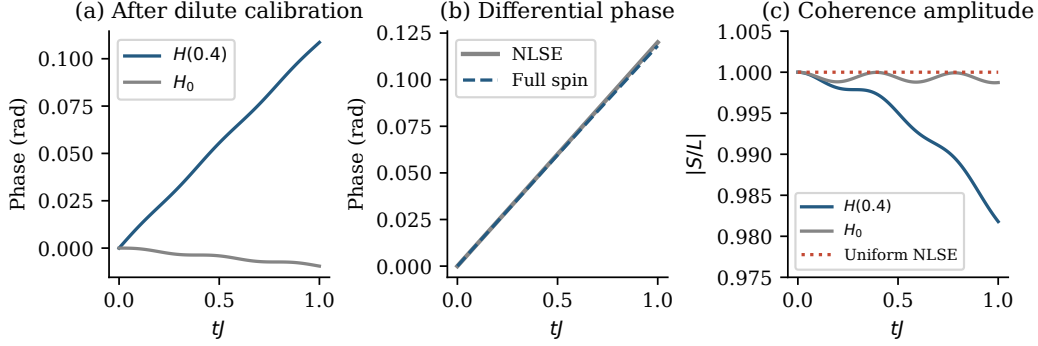


Figure 2: Mechanism-resolving contrast at $\epsilon = 0.4J$. (a) Phase after dilute calibration for the extended and unextended models. (b) Their difference phase agrees with the uniform plasma-envelope phase. (c) The spin coherence amplitude decreases while the uniform classical field maintains its amplitude. Phase agreement therefore does not imply equality of the full evolving states.

The PDE's uniform solution differs from its analytic expression by at most 2.0×10^{-14} over the saved uniform runs. The nonuniform resolution check changes the readout field by at most 1.2×10^{-11} . At $\epsilon = 0.4J$, the production Suzuki step changes the fitted rate by $4.7 \times 10^{-7}J$ relative to the sparse exponential. Thus the percent-scale discrepancy is not explained by the tested time or PDE discretizations. These are deterministic computations; there are no sampling error bars or fitted statistical confidence intervals.

4.2 Recovery of the linear spin-wave model

At exactly $\epsilon = 0$ we separately evolve the extended implementation and an implementation with the ZZ term omitted. Their matrices and statevectors are identical in the dispersion and propagation experiments. On nine sites, a localized vacuum-plus-excitation Ramsey preparation measures all modes for $\Delta = 0$ and $J/4$. An independent nine-dimensional implementation of the first-order logical circuit at the original step $0.8/J$ agrees with the full-statevector fitted frequencies to relative error 4.2×10^{-16} for nonzero modes. The zero mode is handled with an absolute error rather than an undefined relative error.

There is an important difference between reproducing a discrete logical circuit and measuring the continuous Hamiltonian spectrum. At step $0.8/J$, projecting onto continuous eigenmodes produces beating over the finite observation interval. Its fitted slopes differ from the exact Hamiltonian by up to 9.90% at zero mass and 17.68% at $\Delta = J/4$. We retain these values in Appendix B, and disclose a subsequent, separately timestamped step-refinement check. Reducing the Ramsey step to $0.05/J$ gives a maximum error of 0.934% over both spectra; halving it again reduces the error further. This preserves H_0 and the preparation and uses the same physical observation times.

For propagation, we launch $(|1\rangle + |2\rangle)/\sqrt{2}$ in vacuum, against a sharp $\Delta = J$ step beginning at site 5, and through $\Delta_j = J \exp[-(j-5)^2/(2 \cdot 1.2^2)]$. The Gaussian width is an explicit reconstruction choice because it is not numerically specified in the supplied figure. All runs use 512-dimensional vectors and step $0.8/J$. Figure 3 recovers the ballistic vacuum motion and reflection from overdense profiles of the noiseless protocol of Sundar et al. [2026]. The mean-position difference from our independent logical reference is at most 6.2×10^{-15} sites. This is a logical-circuit and qualitative propagation reproduction, not a reconstruction of unavailable native-gate compilation parameters or experimental data. With our literal Pauli convention the packet first turns at the left boundary; no pointwise match to a digitized published curve is asserted.

5 Correlations, spatial resolution and failure of the closure

A simple solvable limit explains why matching the initial nonlinear coefficient is insufficient. Freeze hopping in the positive-band interior but retain both ZZ neighbors and the product preparation. Conditional on a neighbor's occupancy, a spin's frequency shifts by either $\epsilon/2$ or $-\epsilon/2$. Summing

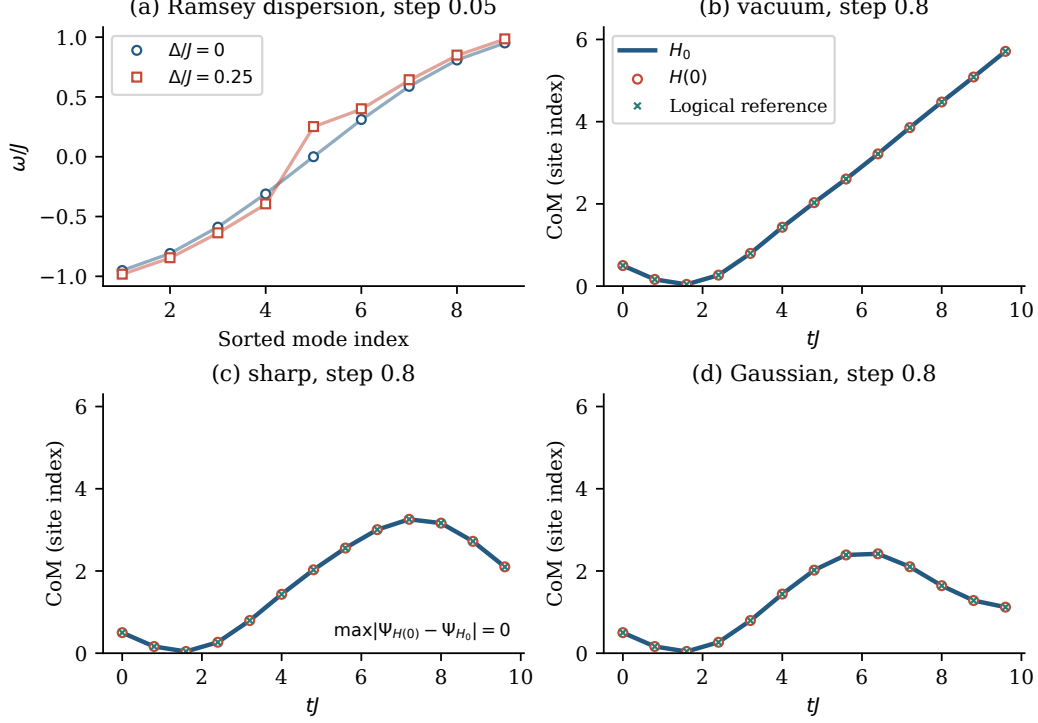


Figure 3: Linear controls at $\epsilon = 0$. (a) Exact continuous frequencies (lines) and refined full-vector Ramsey estimates (symbols) for both masses. (b–d) Unextended H_0 , the ZZ implementation evaluated at zero coupling, and an independently constructed logical circuit have overlapping mean-position trajectories. CoM uses zero-based site coordinates; the Hamiltonian uses one-based labels.

these amplitudes gives the exact local coherence

$$\frac{\langle a_j(t) \rangle}{\langle a_j(0) \rangle} = e^{-i(\Delta + \epsilon)t} (1 - \rho + \rho e^{i\epsilon t})^2. \quad (11)$$

After the vacuum calibration, its phase starts as

$$\Phi_{\text{fr}}(t) = 2\rho\epsilon t - \frac{\rho(1-\rho)(1-2\rho)}{3}(\epsilon t)^3 + \mathcal{O}((\epsilon t)^5). \quad (12)$$

The first term is precisely the plasma NLSE phase. The cubic term is generated by finite-spin occupation fluctuations discarded by factorization. The coherence magnitude also decreases; it equals $1 - 2\rho(1-\rho)[1 - \cos(\epsilon t)]$ in this limit. Connected density-coherence correlations, which enter Eq. (7), are saved directly from the full vectors.

For a continuously sampled fit with an intercept over $[0, T]$, Eq. (12) predicts

$$\frac{R_{\text{fr}} - 2\epsilon\rho}{2\epsilon\rho} = -\frac{3}{20}(1-\rho)(1-2\rho)(\epsilon T)^2 + \mathcal{O}((\epsilon T)^4). \quad (13)$$

The actual fits use the stated discrete samples. Equation (13) estimates a 15% phase-rate boundary near $\epsilon T = 1.30$ for our density. The exact frozen-hopping formula, evaluated without fitting, differs from the measured short-time rates by at most 2.15% across all six couplings. It explains the observed strong-coupling shortfall, but was not used as the PDE reference or to choose the primary interval.

The same model at $\epsilon = 0.4J$, evolved to $T = 8/J$, has 71.74% rate error (Fig. 4). The local formula captures dephasing qualitatively, but accumulating hopping and boundary effects prevent it from remaining an exact description. At long times the quantum model still evolves unitarily; it is the classical closure that fails. This distinction rules out interpreting the short-time result as evidence of sustained soliton or modulational dynamics.

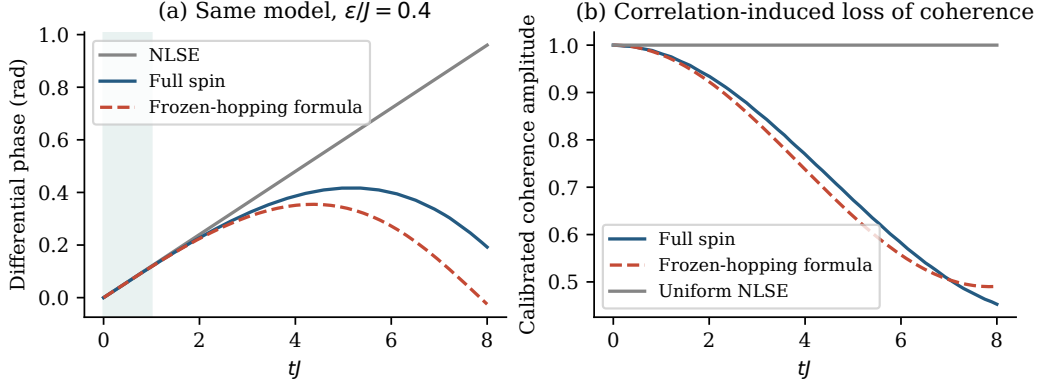


Figure 4: Longer-time failure with the same Hamiltonian, density and coupling as Fig. 2. Shading marks the primary fit interval. The classical phase continues linearly, whereas the quantum phase and coherence show correlation-induced departures. The frozen-hopping expression is an explanatory limit, with no adjusted parameters.

The nonuniform test uses $\rho(x) = 0.15[1 + 0.1 \cos(2\pi x/12)]$, $\epsilon = 0.2J$, and the original interior readout. It gives 3.42% rate error. Appendix D shows all populated sites, including the substantially poorer endpoints. The finite-range density sum samples $\rho(x-2a) + \rho(x+2a)$, so a Fourier modulation is multiplied by $\cos(2ka)$ instead of unity. The endpoints also have half the bulk coordination. These are explicit limitations of a six-cell envelope: the nonuniform test probes local phase response, not a resolved continuum transport solution.

The physical weak-field approximation has an additional, separate limit. At the largest primary coupling, $|\mathcal{A}|^2 = (8\epsilon/\Delta)\rho = 0.06$. Expanding the exact homogeneous relativistic frequency gives a first omitted correction of 3.75% of the cubic shift. This is an analytic estimate of reference-model truncation, not an error measured by solving the full cold-fluid equations. The larger stress couplings further weaken that plasma approximation.

6 Computational cost and relation to other approaches

The propagation algorithm applies local rotations to full vectors in $\mathcal{O}(n_t N 2^N)$ work and $\mathcal{O}(2^N)$ live-state storage. The implementation additionally caches occupation and gate-index arrays in $\mathcal{O}(N 2^N)$ storage. Saving all output times requires $\mathcal{O}(n_{\text{out}} 2^N)$ storage. A 12-spin complex128 vector occupies 64 KiB; a 101-time trajectory occupies 6.31 MiB before compression. The Fourier PDE solver takes $\mathcal{O}(n_t M \log M)$ work and $\mathcal{O}(M)$ live storage. Timings in Appendix D range from approximately 17 to 44 ms for a coherent spin evolution on 9–12 sites; the 128-point PDE evolution at $\epsilon = 0.2J$ takes 5.01 ms. The dilute and zero-coupling controls add evolutions. These single-thread propagation timings exclude observable extraction and I/O and are not asymptotic benchmarks.

Linear plasma-wave spin simulations [Sundar et al., 2026] establish the staggered dispersive backbone. Quantized wave-wave models have also been implemented on superconducting hardware [Shi et al., 2024]; that work targets nonlinear optical mode conversion, whereas our observable isolates the local quartic response of a spatial chain. The classical envelope reduction and spin-to-boson closure are established techniques [Siminos et al., 2014, Holstein and Primakoff, 1940]. The additional content here is the component-local interaction, the explicit removal of its one-body shift and of H_0 saturation, and measured limits on the resulting correspondence.

Developments published after the linear spin-chain study pursue different representations: Berntson et al. [2026] give a weakly nonlinear kinetic algorithm using a controlled Carleman embedding, and Jemcov and Morris [2026] construct a unitary discretization of a Koopman–von Neumann generator. These are complementary approaches, not validations of this ZZ mapping. Our literature survey through October 2026 identified no directly verified extension that supplies the correspondence tested here; this is a bounded search result rather than a priority claim.

An exponential full-vector representation is not evidence of quantum advantage. The unextended chain is a quadratic fermionic problem after Jordan–Wigner transformation [Jordan and Wigner, 1928], and interacting one-dimensional states can remain accessible to matrix-product-state methods [Schollwöck, 2011]. The short-time regime here has weak entanglement and a cheap classical PDE reference. Any advantage argument would require larger, demonstrably difficult correlated states, an appropriate physical quantum-plasma target, state preparation and observable-readout costs, and a comparison with those stronger classical algorithms. None is established by these timings.

7 Conclusions

Component-local ZZ bonds generate the attractive density interaction needed for a weakly relativistic plasma-envelope frequency shift. Full spin evolution passes a fixed differential spectroscopy criterion across the primary coupling range, while preserving the original Hamiltonian exactly at zero coupling. The double calibration is essential, and the observed failures quantify the restricted time and spatial scales over which the correspondence is useful. The results support local nonlinear response calibration on small spin lattices. They do not establish EIT, Raman scattering, modulational growth, nonlinear transport, or the simulation of strongly coupled quantum plasmas.

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A Operator identities and the controlled part of the reduction

With $a_j = |0\rangle_j\langle 1|$ and $n_j = a_j^\dagger a_j$, the on-site identities are

$$[a_j, a_j^\dagger] = 1 - 2n_j, \quad [a_j, n_j] = a_j, \quad [a_j, n_l] = 0 \quad (j \neq l). \quad (14)$$

The chiral exchange in Eq. (1) is

$$H_{\text{hop}} = \frac{iJ}{2} \sum_j (a_j^\dagger a_{j+1} - a_{j+1}^\dagger a_j). \quad (15)$$

Commuting this expression and the density form of Eq. (3) with a_j gives Eq. (7). In particular, the factor $1 - 2n_j$ multiplies hopping, whereas the distinct-site density interaction has no additional on-site saturation factor. Replacing $\langle n_l a_j \rangle$ by $\rho_l \langle a_j \rangle$ is the actual nonlinear closure; substituting a nonlinear Schrödinger equation directly for the quantum state would be a different model.

Define

$$C_{lj} = \langle n_l a_j \rangle - \langle n_l \rangle \langle a_j \rangle. \quad (16)$$

The interaction contribution omitted by factorization is $-\epsilon \sum_{|l-j|=2} C_{lj}$. The stored-state analysis evaluates this complex observable without invoking a mean-field approximation. A density–density connected correlation alone is insufficient: purely diagonal ZZ dynamics can preserve all density probabilities while entangling phases and generating C_{lj} .

In the interior, the density stencil expands as

$$\rho(x - 2a) + \rho(x + 2a) = 2\rho(x) + 4a^2 \rho''(x) + \frac{4a^4}{3} \rho^{(4)}(x) + \dots \quad (17)$$

Consequently the first neglected nonlinear-gradient term is $-4\epsilon a^2 \rho'' \psi$. The bulk linear expansion is

$$E_+(k) = \Delta + \frac{J^2 a^2}{2\Delta} k^2 - \left(\frac{J^2 a^4}{6\Delta} + \frac{J^4 a^4}{8\Delta^3} \right) k^4 + \mathcal{O}(k^6). \quad (18)$$

Both statements are continuum expansions, not equalities on the six-cell chain. Density is measured per component cell; converting a sum to an integral also converts the classical field’s canonical normalization, leaving the local coefficient 2ϵ unchanged.

For comparison with the relativistic plasma, the homogeneous frequency following Eq. (4) at zero wave number is

$$\omega = \Delta(1 + I)^{-1/4} = \Delta \left(1 - \frac{I}{4} + \frac{5I^2}{32} + \dots \right), \quad I = |\mathcal{A}|^2. \quad (19)$$

The identification $I = (8\epsilon/\Delta)\rho$ therefore predicts the shift $-2\epsilon\rho$ at cubic envelope order. Across the primary sweep, I ranges from 0.0075 to 0.06. This is a near-cutoff circularly polarized, weakly relativistic response. It excludes kinetic damping, collisions, mobile-ion feedback, a dynamically resolved density channel and strong-field corrections. In particular, the present spin model is not a derivation of the microscopic electronic–ionic Hamiltonian of a quantum plasma.

Number conservation. Every term in Eq. (2) commutes with $Q = \sum_j n_j$. Hence

$$[H, Q] = 0, \quad \frac{d}{dt} \langle P_m \rangle = 0 \quad (20)$$

for the projector P_m onto each excitation sector. A pure one-excitation preparation remains in that sector for arbitrary ϵ ; it cannot create a nonlinear mean density by leaking into other sectors. We verify this with a separate finite- ϵ statevector run. The coherent preparation instead contains multiple sectors from the outset, and the observable a_j connects adjacent sectors. Its nonlinear phase depends on the evolving conditional coherences between them even though their total weights do not change.

SymPy records the exact three-site commutator and zero-coupling identity, the ZZ density expansion, and both continuum series in `logs/derivation.json` and `logs/derivation.txt`. An independent Kronecker product of standard Pauli matrices agrees to numerical precision with the bit-indexed three-site Hamiltonian after the same scalar vacuum subtraction. This also fixes the sign convention: the literal operator Eq. (1) is used throughout. Particle–hole or reversed chirality conventions can change displayed frequency signs and the initial packet direction; they are not silently substituted in the comparison.

B Complete linear spectra and propagation specification

The single-excitation block used only as a linear reference is

$$(h_0)_{jj} = -\Delta_j(-1)^j, \quad (h_0)_{j,j+1} = iJ/2, \quad (h_0)_{j+1,j} = -iJ/2. \quad (21)$$

For nine sites and uniform mass its spectrum consists of the four pairs

$$\omega_{l,\pm} = \pm\sqrt{\Delta^2 + J^2 \cos^2(l\pi/10)}, \quad l = 1, \dots, 4, \quad (22)$$

and the unpaired eigenvalue $+\Delta$. These values are evaluated from the equation at runtime, not stored as reference literals. The complete estimates appear in Table 2.

Table 2: All nine frequencies at both masses. The original-step estimator and the smaller-step estimator use the same output times $tJ = 0, 0.8, \dots, 9.6$. Exact values are computed from the open-chain formula. The full 512-amplitude and independent logical-circuit estimates at step $0.8/J$ agree to roundoff, but those slopes are not exact continuous-Hamiltonian eigenvalues. The refined nonzero-mode error is relative to the exact frequency; the zero mode is checked absolutely.

Δ/J	Exact ω/J	Ramsey, $\delta = 0.8$	Ramsey, $\delta = 0.05$	Refined error (%)
0	-0.951057	-0.953935	-0.951510	0.048
0	-0.809017	-0.800356	-0.808905	0.014
0	-0.587785	-0.595385	-0.588631	0.144
0	-0.309017	-0.339598	-0.311195	0.705
0	0.000000	-0.000000	-0.000000	$ \delta\omega < 10^{-14} J$
0	0.309017	0.339598	0.311195	0.705
0	0.587785	0.595385	0.588631	0.144
0	0.809017	0.800356	0.808905	0.014
0	0.951057	0.953935	0.951510	0.048
0.25	-0.983366	-0.949290	-0.981857	0.153
0.25	-0.846764	-0.798192	-0.844542	0.262
0.25	-0.638742	-0.579103	-0.636322	0.379
0.25	-0.397481	-0.327213	-0.394200	0.825
0.25	0.250000	0.274851	0.252257	0.903
0.25	0.397481	0.454244	0.401196	0.934
0.25	0.638742	0.719275	0.644147	0.846
0.25	0.846764	0.884743	0.850045	0.388
0.25	0.983366	1.015166	0.985651	0.232

The full-vector Ramsey state is $(|\text{vac}\rangle + |1\rangle)/\sqrt{2}$. With the vacuum scalar removed, $2\langle a_j(t) \rangle$ equals the single-excitation propagation amplitude. We project it on each exact eigenvector of h_0 , unwrap its phase relative to $t = 0$, and take minus its fitted slope. For the first-order circuit, $U_\delta = U_D(\delta)U_E(\delta)U_O(\delta)$, where O and E contain alternating exchange bonds and D contains the diagonal field. A continuous eigenvector is generally not a Floquet eigenvector of this product, causing the finite-window beating reported in the main text. Independent Floquet frequencies are also logged. The refinement to $0.05/J$ and $0.025/J$ was added after diagnosing that issue; neither the original-step records nor the nonlinear analysis was replaced.

For propagation, the barrier arrays are those in Sec. 4.2; all boundaries remain open. Populations, coherences, full statevectors and centers of mass at every output time are saved. Vacuum, sharp-step and Gaussian continuous-time sparse-exponential controls are additionally supplied. No experimental noise, shots, mitigation or undocumented native-gate compilation is included. The Gaussian reconstruction and the initial reflection in our Pauli convention limit how literally these curves can be overlaid with the published image; the verified features are free propagation followed by finite-boundary effects, and earlier reflection from either overdense profile.

C Numerical accuracy and sector-resolved checks

The production symmetric product is

$$U_2(\delta) = e^{-iH_D\delta/2}e^{-iH_O\delta/2}e^{-iH_E\delta}e^{-iH_O\delta/2}e^{-iH_D\delta/2}, \quad (23)$$

with all ZZ bonds and staggered fields in H_D . They commute and are applied as one diagonal phase. Exchange gates rotate every pair of basis amplitudes differing by a neighboring $10 \leftrightarrow 01$ swap; amplitudes 00 and 11 are unchanged by that exchange bond. The implementation iterates over the complete binary index array, with site j stored in bit $j - 1$. It never selects an excitation sector for propagation. The independent sparse exponential uses the same full-dimensional Hamiltonian assembled in COO/CSR format, with its scalar vacuum energy removed.

Table 3 shows the fitted observable’s convergence at the largest primary coupling. The statevector norm error across the primary sweep is at most 4.7×10^{-14} , and the maximum sector-weight drift is 2.3×10^{-14} . For the pure one-excitation control at $\epsilon = 0.4J$, population outside that sector is exactly zero; the sparse number commutator is also exactly zero. These tests would reveal spurious sector creation, not establish physical conversion between conserved sectors.

Table 3: Independent time-discretization check at $N = 12$, $\epsilon = 0.4J$, $\rho = 0.15$. Every value is extracted after the same dilute and H_0 calibrations. The reference uses full-dimensional sparse matrix-exponential actions for both the signal and controls.

δJ	R_{spin}/J	$ R - R_{\text{exp}} /J$
0.01	0.118238649	1.87×10^{-6}
0.005	0.118240051	4.67×10^{-7}
0.0025	0.118240401	1.17×10^{-7}
exponential	0.118240518	0

The coherent state’s initial sector probabilities follow the binomial distribution of the six active spins and are explicitly measured from the vector. Weight in $m \geq 2$ remains 22.35%, while the configuration probabilities within those sectors change. At $\epsilon = 0.4J$, half the L^1 change in these configuration probabilities at $tJ = 1$ is 1.4×10^{-3} . The maximum single-site linear entropy during that run is 0.00894, confirming departure from the initial product state. Phase correlations can grow even when these population changes are small. All 2^{12} amplitudes are retained; the fact that initially empty conserved sectors remain empty is a selection rule rather than a computational truncation.

The NLSE is integrated by symmetric kinetic/nonlinear splitting. With $k = 2\pi n/L$, the kinetic half-step multiplies each Fourier amplitude by $\exp(-iDk^2\delta/2)$; the nonlinear step multiplies each grid value by $\exp(2i\epsilon|\psi|^2\delta)$. Readout at physical positions $x = 4a, 6a$ uses Fourier interpolation, including when these positions do not fall exactly on grid points. PDE rates use the same sample times and least-squares definition as the spin rates. The nonuniform zero-coupling PDE field supplies the linear denominator in that case. All uniform fields, nonuniform fields and convergence results were computed and saved independently of the quantum trajectories.

D Additional validation and computational cost

Table 4 includes every additional case specified with the original method. Varying the density checks that the incremental phase is proportional to population, not merely a coupling-dependent offset. The size controls use the same physical readout sites and mass, preparing every available odd site. Their agreement does not constitute a spatial continuum convergence test: the spacing is held fixed and the number of cells remains small.

Figure 5 displays the phase at every populated spin for the modulated input. The readout sites were specified before evolution. Endpoint disagreements are retained, rather than selecting an interior region after seeing the data. The wave number of the density modulation is $k = 2\pi/(12a)$, for which the interaction stencil multiplies the modulation by $\cos(2ka) = 1/2$. This explains why a good central average need not imply a quantitatively accurate full spatial profile. At the two ends, the local phase has only one interacting neighbor. Dilute calibration removes their one-body potential difference but cannot supply the missing nonlinear bond. Boundary engineering or a much finer envelope lattice would therefore be needed before comparing spatial nonlinear wave propagation.

Full statevectors, reference fields and provenance records accompany this report. `code/run_all.sh` regenerates the calculations; `code/build_paper.sh` compiles and renders the manuscript. The reproduction guide and implementation/attempt logs document source attribution, the fixed protocol and failed regimes.

Table 4: All additional checks of the fixed ZZ model. Except for the long-time row, the fit uses $0 \leq tJ \leq 1$. The density and size rows share the stated uniform PDE reference. These are deterministic controls; repeated parameter combinations check the same model and are not independent samples.

Control	N	R_{spin}/J	R_{PDE}/J	Error (%)
$T = 8/J, \epsilon = 0.4J$	12	0.033906	0.120000	71.74
$\rho = 0.05, \epsilon = 0.2J$	12	0.019866	0.020000	0.67
$\rho = 0.15, \epsilon = 0.2J$	12	0.059639	0.060000	0.60
$\rho = 0.3, \epsilon = 0.2J$	12	0.119372	0.120000	0.52
size, $\epsilon = 0.2J$	9	0.059647	0.060000	0.59
size, $\epsilon = 0.2J$	10	0.059601	0.060000	0.66
size, $\epsilon = 0.2J$	11	0.059638	0.060000	0.60
size, $\epsilon = 0.2J$	12	0.059639	0.060000	0.60
10% modulation, $\epsilon = 0.2J$	12	0.057416	0.055519	3.42

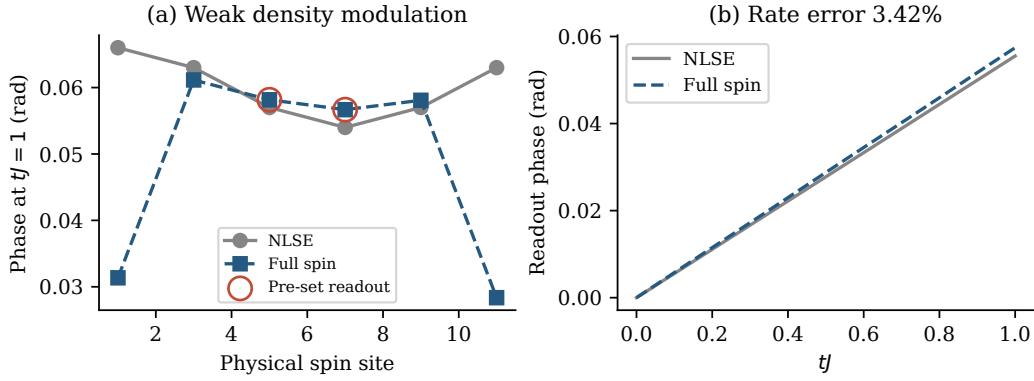


Figure 5: Weakly modulated input at $\epsilon = 0.2J$. (a) Every active site's differential phase at $tJ = 1$, including both open endpoints. Red rings indicate the fixed interior readout. (b) The mean readout phase used for the 3.42% frequency comparison. Spatial agreement is visibly poorer than agreement of that scalar observable.

Table 5: Measured propagation-only wall times, using one numerical-library thread, 200 Suzuki steps and 101 stored output times. All rows use $\epsilon = 0.2J$, $\Delta = 8J$, $\rho = 0.15$ and the same graph. The 128-point PDE reference takes 5.01 ms. These are single-run measurements, so timer/scheduler variation limits close comparisons.

N	Dimension	Coherent evolution (ms)	Dilute calibration (ms)
9	512	16.79	16.79
10	1024	21.65	21.28
11	2048	30.55	30.29
12	4096	44.02	47.64